Representation Effects in Indic Factual Question Answering: Script, Output Budget, and Judge Dependence

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***Abstract*—Cross-lingual factuality comparisons usually change a question's content or phrasing along with its language, so a drop in accuracy has no single cause. We hold the proposition fixed and vary only its representation. We build paired prompts for 100 propositions on 20 Indian public-policy topics in five Indic languages (Hindi, Bengali, Tamil, Telugu, Kannada), each rendered in English, native script, Romanized text, Latin-script code-switching (CS-LAT) and mixed-script code-switching (MIXED); in Hindi, Bengali and Tamil the last two differ only in the script of the Indic words. With Qwen3.8-27B (API identifier qwen/qwen3.8-27b), greedy decoding, a 128-token output cap and one judge configuration for all conditions, judge-assessed accuracy is 60% for a template-matched English prompt and 49%, 37%, 32% and 26% for CS-LAT, Romanized, native-script and MIXED prompts; MIXED is 23 points below CS-LAT (exact McNemar $p < 0.001$; topic-level permutation $p = 0.004$). Two follow-up checks change how this should be read. Every response reached the cap, and Brahmic-script prompts elicit Brahmic-script answers that fit about half as many characters into it; with a 512-token cap the CS-LAT–MIXED gap disappears (+4 points, $p = 0.57$). A judge from another model family agrees with the self-judge on 76% of outputs and finds no significant CS-LAT–MIXED gap even at 128 tokens. The English–CS-LAT gap is not significant at the topic level in any analysis. For this model and benchmark, the apparent cost of script alternation is mainly an output-budget effect; representation effects should be measured with non-binding output budgets and more than one judge. Prompts, outputs, judgements, a gold-answer ledger and analysis code are released.**

***Index Terms*—factual question answering, code-switching, script, Indic languages, output length, LLM-as-a-judge, paired evaluation, cluster-robust inference**

I. INTRODUCTION

A farmer who wants to know whether crop insurance under PMFBY pays out on measured yield loss or on a weather index can type that question in at least five ways: in English; entirely in Devanagari; in Latin-script Hindi; in the Hindi–English mixture people actually text, still in Latin script; or in the same mixture with Hindi particles in Devanagari and the scheme names and terms such as *main rule* left in Latin letters. All five ask for the same fact. If a model answers some of them correctly and others not, what changed? The knowledge is the same, and so is the target answer. What differs is the surface form.

Existing multilingual evaluations rarely isolate this variable. Many benchmarks translate or re-author questions per language, so content, difficulty and phrasing move together with the language [1], [2], [3]. Code-switching benchmarks mostly measure task performance on naturally occurring text rather than recall of a fixed proposition [4]. Work on tokenization shows that some languages cost more tokens per unit of content [5], [6], [7], but that is usually inferred from corpus-level correlations, not from controlled prompt pairs. A fall in accuracy for "Indian-language prompts" could therefore reflect missing knowledge, harder questions, Romanization, the presence of English insertions, script, or tokenization, and these cannot be separated after the fact.

We build the comparison so that they can be. Each of 100 propositions is rendered in five paired representations (Table I). Two of them, CS-LAT and MIXED, are meant to differ only in whether the Indic words are written in Latin or Brahmic script; this holds word for word in Hindi, Bengali and Tamil, while in Telugu and Kannada the question word also differs (Section III-A). We add a second English condition after finding that the original English template was more specific than the template shared by the other four conditions (Section III-B). We then ask whether the differences we measure survive two checks that representation studies rarely run: a larger output budget, and a judge from a different model family.

TABLE I

One proposition in all six conditions (Hindi). In Hindi, CS-LAT and MIXED contain the same words in the same order and differ only in the script of *ke, ya, kya hai* (also true for Bengali and Tamil; in Telugu and Kannada the question word differs, Section III-A). The topic string stays in Latin script in every condition, so NATIVE contains one Latin–Brahmic boundary per prompt.

Condition	Prompt (topic: PMFBY vs. WBCIS, Hindi group)	Script of Indic words
EN-ORIG	What is the key structural or operational difference between PMFBY vs WBCIS?	(English)
EN-MATCH	What is the main rule, date or parameter regarding PMFBY vs WBCIS?	(English)
NATIVE	PMFBY vs WBCIS के संबंध में मुख्य नियम, तिथि या मापदंड क्या है ?	Devanagari
ROMAN	PMFBY vs WBCIS ke sambandh mein mukhya niyam, tareekh ya mapdand kya hai?	Latin
CS-LAT	PMFBY vs WBCIS ke regarding main rule, date ya parameter kya hai?	Latin
MIXED	PMFBY vs WBCIS के regarding main rule, date या parameter क्या है ?	Devanagari

Research questions. RQ1: does judge-assessed answer accuracy depend on representation when the proposition is held fixed? RQ2: does mixed-script code-switching lower accuracy relative to the same code-switched words in Latin script? RQ3: does the effect differ across the five languages? RQ4: are the effects robust to the output budget and to the choice of judge? RQ5: is the script effect associated with input tokenization length? Hallucination detectors and mitigation are outside the scope of this paper.

Contributions. (1) A paired six-condition design for Indic factual QA, with a matched-English control that removes a template confound we found in our first run. (2) Evidence that a binding output budget can manufacture a script effect: the 23-point MIXED penalty measured at 128 tokens disappears when the same prompts are answered with 512 tokens. (3) Evidence of judge dependence: grading all conditions with one judge configuration, rather than the two used in our first run, shrinks the English–code-switching gap from 17 to 11 points, and a cross-family judge changes which contrasts are significant. (4) Inference that treats the 20 topics, not the 500 prompts, as the unit of independent evidence, and a release of prompts, outputs, judgements, a gold-answer ledger, and protocols for the human validation that is still outstanding.

Findings in brief. At a 128-token budget, accuracy falls from English to the Indic representations, and mixed script is lowest; under the self-judge the MIXED–CS-LAT gap is significant at prompt, proposition and topic level. It is not robust. Almost every response hits the cap, Brahmic-script answers are about half as long in characters, and with 512 tokens MIXED reaches CS-LAT (63% vs. 59%) and the level English reached with 128 tokens. A cross-family judge is much stricter (37% vs. 60% on English) and finds no significant MIXED–CS-LAT gap at either budget. The English advantage over Latin-script code-switching is not significant at the topic level in any analysis. Input characters per token, measured with the served model's tokenizer, shows no detectable association with correctness within the controlled contrast.

II. RELATED WORK

Code-switching and LLMs. GLUECoS evaluates code-switched understanding across tasks and language pairs including Hindi–English [4]. More recent studies find that large multilingual models are unreliable on code-switched tasks and when asked to generate code-mixed text [8], [9]. These works vary the task or the generated text; we hold the proposition fixed and vary only its rendering.

Multilingual evaluation for Indic languages. IndicNLPSuite and IndicXTREME provide corpora and task benchmarks for Indic languages [2], [3], and MEGA compares generative models across many languages and tasks [1]. Such benchmarks answer how well a model performs per language, but their per-language items are not the same questions. Romanized input has itself been proposed as a bridge to Indic languages for LLMs [23]; our ROMAN condition measures the same factor on fixed propositions.

Tokenization. Subword vocabularies fragment low-resource scripts into more tokens [7], [10], with measurable differences in cost and quality across languages [5], [6]. The same premium applies to generated text: an answer in a script that costs more tokens per character fits less content into a fixed output limit, which is the mechanism our budget experiment isolates. Our mediation analysis asks whether the input-side quantity explains the script contrast.

Factuality evaluation and LLM judges. We use closed-book questions with a gold answer, closer in spirit to TruthfulQA-style checks than to long-form factual precision scoring [11], [12]. LLM graders are convenient but biased, for example towards their own outputs, answer position and length [13], [24], [25]; we therefore treat the judge as a central threat and grade with two judges from different model families.

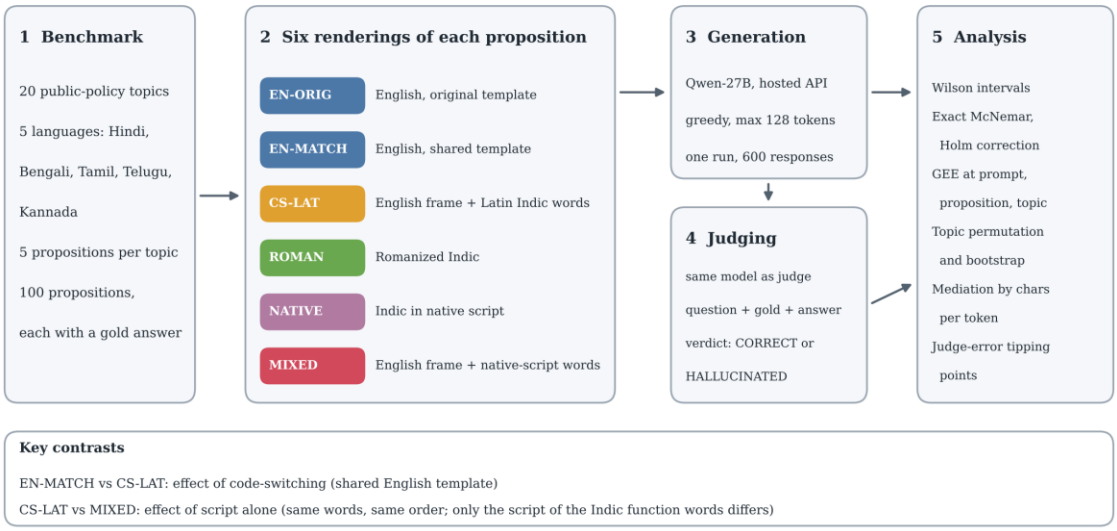


Fig. 1. Study design. Each proposition is rendered in six conditions and answered once by Qwen3.8-27B under greedy decoding with a 128-token limit; responses are graded against the gold answer by the same model (J-QWEN). The follow-up (Section VI-B) re-answers three conditions with a 512-token limit and grades both budgets with J-QWEN and with gpt-oss-120b (J-OSS). Key contrasts: EN-MATCH vs. CS-LAT (code-switching) and CS-LAT vs. MIXED (script).

III. BENCHMARK AND PAIRED DESIGN

A. Propositions, languages and conditions

The evaluated core contains 100 semantic groups: 20 topics, each instantiated once per language in Hindi (hi), Bengali (bn), Tamil (ta), Telugu (te) and Kannada (kn). A group has one gold answer and evidence snippet, shared by its prompts in all conditions. The 20 topics are Indian public-policy and regulatory subjects (for example PMFBY vs. WBCIS, NEFT vs. RTGS, RTI third-party information, Companies Act CSR thresholds, IBC Section 7 default threshold, Citizenship Amendment Act cut-off); the full list is in Appendix A. Topics are crossed with languages, so language effects are not confounded with topic.

Table I defines the six conditions. In NATIVE, ROMAN, CS-LAT and MIXED the question frame is one fixed template ("main rule, date or parameter regarding *X*") with *X* the topic string; only the rendering varies. ROMAN and NATIVE render the same Indic frame in Latin and Brahmic script, except in Hindi, where ROMAN uses *tareekh* and NATIVE *tithi* for "date". CS-LAT replaces part of the frame with English words ("main rule", "date", "parameter", and in Hindi and Bengali also "regarding"), and MIXED writes the remaining Indic words of CS-LAT in native script. In Hindi, Bengali and Tamil, CS-LAT and MIXED are word-for-word transliterations of each other. In Telugu and Kannada they are not: the question word is *enti* vs. □□□□□ (*emiti*) in Telugu and *yenu* vs. □□□□□□ (*yāvudu*, "which") in Kannada. We report the CS-LAT–MIXED contrast on all 100 propositions, as planned, and on the 60 word-identical ones as a sensitivity analysis. No fluent speaker has yet rated the naturalness or equivalence of these templates; the templates and a review protocol are released.

B. Amendment: matched English

In the first run (EXP-002) the English prompts used two question templates ("key structural or operational difference between *X*" and "specific prerequisite condition or timeline applies to *X*"), while all other conditions used the single generic template. English-versus-Indic comparisons were therefore confounded with question type, a flaw we found only when auditing the raw prompt strings. We added EN-MATCH, the generic template in English (100 prompts, same gold answers), and ran it in a second run (EXP-003) with the same model and decoding settings, without modifying the earlier results. The two runs did not grade alike: EXP-002 called the judge at temperature 0.3 with the answering system prompt, EXP-003 at temperature 0 with an evaluator system prompt (Section IV). We therefore re-graded all 600 outputs with the EXP-003 configuration; all comparisons below use these harmonised labels and EN-MATCH as the English reference. EN-ORIG and EN-MATCH do not differ detectably (70% vs. 60%; exact $p = 0.10$, topic-level $p = 0.69$).

Because the English prompt is identical across the five language tags of a topic and decoding is greedy, English outcomes are near-duplicated within topic: in EN-MATCH, 12 topics are correct in all five groups and 8 in none. The English condition therefore carries about 20 independent observations, not 100, which is why we report topic-level bootstrap intervals next to Wilson intervals.

C. Provenance, and an earlier benchmark that was retired

The core is the out-of-distribution test partition of IndraLLM-CS v1.1-CANDIDATE (1,500 semantic groups, 7,500 prompts; development, validation, in-distribution and out-of-distribution partitions, twelve template families). Version 1.0 (2,000 groups, 10,000 prompts) was retired after an internal audit showed that train and test partitions shared

prompts, answers, entity pairs and templates at every level checked, so it supports no generalization claim. The rebuilt benchmark separates partitions by template family, exact and near-duplicate text, evidence and entity pair. The 1,000 in-distribution prompts per model that were also run (EXP-002) contain fictitious entities, and the model answered 2 of 2000 correctly across both models; they carry no information about recall and are excluded here (Appendix A). A further 25 propositions (version 1.2-PILOT) have been written but **have not been evaluated by any model**; we use them only for a prospective design calculation (Appendix D).

Gold answers. Each topic has one gold answer, shared by its five language groups and all six conditions. The benchmark stores an evidence snippet and a source URL with each; the snippets are template-generated paraphrases rather than quotations, and the URLs point to agency home pages (e.g. rbi.org.in) rather than to specific documents. The author and, separately, volunteer reviewers known to the author checked the gold answers for factual correctness against public sources. That check was not logged item by item, the reviewers were not trained annotators, and no agreement statistic was computed, so we report it as a check rather than a measurement. For this version we also matched each gold answer to its governing provision and reviewed it for currency and for completeness relative to the question actually asked (Appendix A, Table VIII; a per-topic ledger is released). All 20 answers are consistent with the provision identified. One describes a scheme that has since been replaced (Telangana's Rythu Bandhu, now Rythu Bharosa). Three are narrower than the generic question allows, so that other correct answers exist: the GST registration threshold (₹40 lakh applies to suppliers of goods, ₹20 lakh to services), the MTP Act (two practitioners are required only between 20 and 24 weeks) and the environmental-clearance public hearing (the gold lists two of several exemptions). Removing these three topics changes no conclusion (Appendix B). Gold-answer checks establish that the reference is right; they do not show that the judge grades model answers correctly, which no human has yet assessed on these outputs (Section VII). Agreement figures in older project documents (e.g. $\kappa = 0.719$) concern a different, earlier pilot whose annotation files are not in the repository, and are not used here.

IV. EXPERIMENTAL SETUP

Fig. 1 summarizes the pipeline: 100 propositions are rendered in six conditions, answered once by the primary model under greedy decoding, graded against the gold answer, and analysed at three levels of clustering. Section VI-B adds a follow-up run with a larger output budget.

Models. The primary model is served by Groq under the API identifier `qwen/qwen3.8-27b`, which is recorded in every prediction row and in the response metadata of the follow-up. The provider lists it as an Alibaba Cloud model, and a public checkpoint of that name exists (Qwen/Qwen3.8-27B, 27.8B parameters, Apache-2.0). That checkpoint's tokenizer reproduces the API's prompt-token counts for all 600 prompts up to a constant chat-template offset of 48 tokens, and counts 125–128 tokens for responses the API reports as 128, so the tokenizer is established; whether the provider serves exactly those weights, and at what numerical precision, we cannot verify. We call the model Qwen3.8-27B. Older project documents call it "Qwen-2.5-27B" and the repository configuration lists `qwen/qwen3.6-27b`; neither produced the reported outputs. The second model is ALLaM-2-7B (`allam-2-7b`) [14]. All results are single-run. Greedy decoding on the hosted API was not fully deterministic: in the follow-up, 84–85% of 512-token responses begin with exactly the 128-token response to the same prompt, and the rest diverge.

Prompting and output budget. Each prompt is sent with the system message "You are a helpful assistant answering questions from Indian users. Questions may mix an Indian language with English. Answer factually, accurately, and concisely." Decoding is greedy (temperature 0, top-p 1) with at most 128 output tokens in the main runs. The model almost never stops on its own: 598 of 600 responses used all 128 tokens, and 82–95% end mid-sentence in every condition, English included. The provider's finish reason was not logged in the main runs; it was logged in the follow-up.

Judging. A judge receives the question as the model saw it, the gold answer and the response, and returns CORRECT or HALLUCINATED with a short reason (Appendix C). Three label sets exist. *Stored*: the labels saved with the predictions; EXP-002 obtained them from Qwen3.8-27B at temperature 0.3 with the answering system prompt (a side effect of reusing the generation client), EXP-003 at temperature 0 with the system prompt "You are an expert factuality evaluator." *J-QWEN*: all 600 outputs re-graded by Qwen3.8-27B with the EXP-003 configuration, the one the study intended; these labels are the primary analysis. *J-OSS*: the same rubric applied by gpt-oss-120b (`openai/gpt-oss-120b`, temperature 0, low reasoning effort), a judge from a different model family that did not produce the answers. J-QWEN reproduces the stored labels on all 100 EN-MATCH items (same configuration) and on 90–94% of items in the other conditions. No judge has been validated against human labels on these outputs. "Accuracy" below is the share of responses a judge labels CORRECT: a judge-assessed quantity, not a direct measurement of recall.

Statistics. Observations are nested. A *prompt* is one string sent to the model. A *proposition* is a fact in one language group, rendered as six paired prompts. A *topic* (e.g. NEFT vs. RTGS) contains five propositions, one per language, with the same gold answer; across the five matched-template conditions it contributes 25 prompts. A proposition-level test asks whether the model, given these 100 facts, does better in one representation than in another. A topic-level test asks whether the difference should be expected to recur on new topics; with 20 topics it has much less power, but it matches the generalisation most readers want, so we report both. At the proposition level we use exact McNemar tests [15] with Holm correction [16] within a pre-specified family of five contrasts: each of NATIVE, ROMAN, CS-LAT and MIXED against EN-MATCH, and MIXED against CS-LAT. At the topic level we use an exact sign-flip permutation test over all 2^{20} sign patterns of the 20 per-topic accuracy differences and a percentile bootstrap over topics (10,000 resamples, fixed seed), with Holm correction within the same family. Logistic GEE with

independence working correlation and robust covariance clustered at prompt, proposition and topic level [18] is reported for comparison; with 20 clusters robust standard errors are anti-conservative [19], [20], so topic-level conclusions do not rest on it. Wilson intervals [17] are given for single proportions; the design effect is $1 + (m - 1)\rho$ with ρ from the one-way ANOVA estimator [21]. The follow-up (Section VI-B) has three contrasts, fixed in the analysis script before its outputs were graded: CS-LAT vs. EN-MATCH and MIXED vs. CS-LAT at 512 tokens, and the change in each gap from 128 to 512 tokens. Everything else is exploratory. All numbers are recomputed from the raw files by the scripts in Appendix E.

V. RESULTS AT A 128-TOKEN BUDGET

A. Accuracy falls with representation

Table II and Fig. 2 give the main result under the harmonised judge. Relative to EN-MATCH (60%), accuracy is 49% for CS-LAT, 37% for ROMAN, 32% for NATIVE and 26% for MIXED, differences of 11, 23, 28 and 34 points. At the proposition level the ROMAN, NATIVE and MIXED differences are significant after Holm correction (adjusted $p \leq 0.002$); the CS-LAT difference is not ($p = 0.13$). With the stored labels, which graded EN-MATCH and the Indic conditions under different judge settings, the CS-LAT gap was 17 points ($p = 0.014$): about a third of it came from the grading difference. The ordering among the three non-English-script conditions is uncertain: ROMAN vs. MIXED gives $p = 0.052$ and NATIVE vs. MIXED $p = 0.29$ (Table IX).

TABLE II

Qwen3.8-27B accuracy on the 100 paired propositions at a 128-token budget, harmonised judge (J-QWEN). Δ: difference in correct answers out of 100 relative to EN-MATCH. Exact p: two-sided exact McNemar; Holm: within the five primary contrasts (MIXED vs. CS-LAT is in the text and Table IX). Topic p: exact sign-flip permutation test on the 20 per-topic differences, Holm-adjusted in parentheses. Stored, J-OSS: accuracy (%) under the original labels and under the cross-family judge. †EN-ORIG vs. EN-MATCH, exploratory.

Condition	Acc. (%)	Wilson CI	Topic-boot. CI	Δ	Exact p	Holm p	Topic p (Holm)	Stored	J-OSS
EN-ORIG	70.0 (70/100)	[60.4, 78.1]	[50, 90]	+10†	0.10	–	0.69	64	40
EN-MATCH	60.0 (60/100)	[50.2, 69.1]	[40, 80]	ref.	–	–	–	60	37
CS-LAT	49.0 (49/100)	[39.4, 58.7]	[34, 64]	−11	0.13	0.13	0.42 (0.42)	43	25
ROMAN	37.0 (37/100)	[28.2, 46.8]	[24, 50]	−23	0.001	0.002	0.09 (0.17)	33	24
NATIVE	32.0 (32/100)	[23.7, 41.7]	[18, 46]	−28	2.54×10^{-5}	1.02×10^{-4}	0.03 (0.12)	28	21
MIXED	26.0 (26/100)	[18.4, 35.4]	[12, 41]	−34	5.38×10^{-6}	2.69×10^{-5}	0.03 (0.12)	24	17

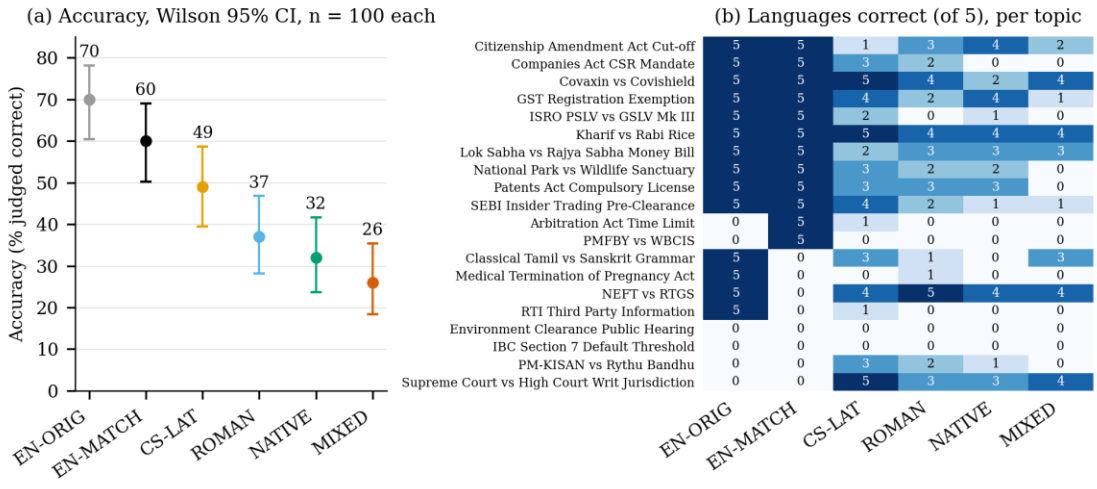


Fig. 2. (a) Accuracy (J-QWEN) with Wilson 95% intervals (n = 100 per condition; points not independent within topic). (b) Number of languages (of five) in which each topic is answered correctly, per condition. Topics sorted by EN-MATCH, then EN-ORIG.

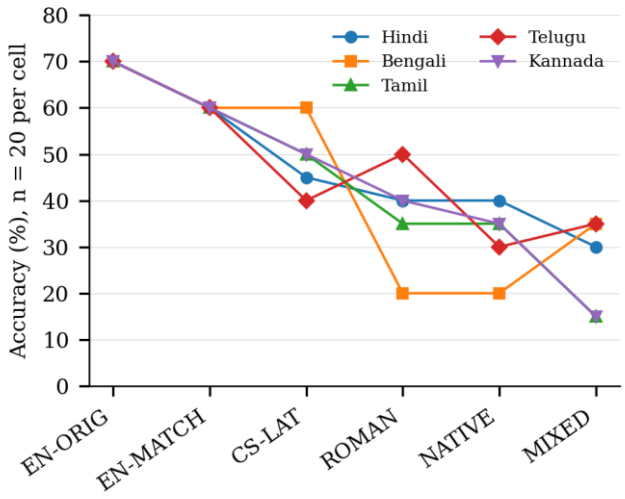


Fig. 3. Accuracy by language and condition (J-QWEN; n = 20 per cell).

The uncertainty is larger than the Wilson intervals imply. Topic-bootstrap intervals are wider (for EN-MATCH [40, 80] against Wilson [50.2, 69.1]) because accuracy is concentrated by topic: two topics (Environment Clearance Public Hearing, IBC Section 7) are wrong in every condition, and twelve are right for English in all five languages (Fig. 2b). The two all-wrong topics fail for different reasons. For IBC the model gives the pre-2020 default threshold of ₹1 lakh, a genuine error; for Environment Clearance the gold answer lists exemptions that the generic question does not ask for (Section III-C).

B. Mixed vs. Latin-script code-switching

Writing the code-switched prompt with its Indic words in Brahmic rather than Latin script lowers accuracy from 49% to 26%, a 23-point drop. Of 37 propositions that differ, 30 favour CS-LAT and 7 favour MIXED (exact $p < 0.001$, Holm $p < 0.001$). The drop appears in all five languages (Fig. 3); the mean of the 20 per-topic differences is 23 points (bootstrap 95% CI [11, 35]; permutation $p = 0.004$, Holm-adjusted 0.018; 14 of 20 topics differ), and the GEE odds ratio is 0.37 (topic-clustered 95% CI 0.21–0.65). On the 60 propositions in Hindi, Bengali and Tamil, where the two prompts are word-for-word identical, the drop is 25 points ($p = 0.0015$; topic-level $p = 0.016$). Excluding the three topics with narrow gold answers leaves it at 23.5 points ($p < 0.001$). Under the stored labels it was 19 points ($p = 0.003$; topic-level $p = 0.015$, Holm-adjusted 0.072). At a 128-token budget, then, the mixed-script penalty is the most consistent result in the study.

TABLE III

Logistic GEE (independence working correlation, robust covariance), J-QWEN labels, 128 tokens. $\hat{\beta}$ is the log-odds difference from EN-MATCH (first four blocks, 500 observations) or from CS-LAT (last block, 200). Topic-level robust SEs with 20 clusters are anti-conservative.

CS-LAT vs. EN-MATCH	**prompt (500)**	**−0.445**	**0.286**	**0.12**
	proposition (100)	−0.445	0.264	0.09
	topic (20)	−0.445	0.493	0.37
ROMAN vs. EN-MATCH	prompt (500)	−0.938	0.291	0.001
	proposition (100)	−0.938	0.273	5.95×10^{-4}
	topic (20)	−0.938	0.494	0.06
NATIVE vs. EN-MATCH	prompt (500)	−1.159	0.296	9.00×10^{-5}
	proposition (100)	−1.159	0.263	1.06×10^{-5}
	topic (20)	−1.159	0.473	0.01
MIXED vs. EN-MATCH	prompt (500)	−1.451	0.306	2.10×10^{-6}
	proposition (100)	−1.451	0.310	2.89×10^{-6}
	topic (20)	−1.451	0.606	0.02
MIXED vs. CS-LAT	prompt (200)	−1.006	0.303	9.11×10^{-4}
	proposition (100)	−1.006	0.257	8.96×10^{-5}
	topic (20)	−1.006	0.293	5.88×10^{-4}

Section VI shows that this result does not survive a larger output budget. Here we note only why that check is needed: 88% of MIXED responses are written mostly in Brahmic script, against 2% of CS-LAT responses, so the prompt script also changes the script of the answer, and with it how much answer fits into 128 tokens.

C. Clustering changes the strength of the English–code-switching gap

Observations within a topic are strongly dependent. In the five matched-template conditions the intra-topic correlation is $\rho = 0.27$, the design effect 7.5 and the effective sample size 67 out of 500. Table III and Fig. 4a show what this does. For CS-LAT against EN-MATCH, the GEE p value is 0.12 at prompt level, 0.09 at proposition level and 0.37 at topic level, and the topic-level permutation test gives $p = 0.42$ (bootstrap 95% CI for the mean per-topic difference [−34, 13] points). For ROMAN, NATIVE and MIXED against EN-MATCH the topic-level permutation p values are 0.087, 0.031 and 0.030, and none survives Holm correction at topic level (smallest adjusted value 0.12). Only MIXED vs. CS-LAT survives Holm correction at every level (adjusted topic-level $p = 0.018$). The English-vs-Indic differences are consistent in direction, but at the level of topics only the larger ones approach significance, and the English–CS-LAT gap does not.

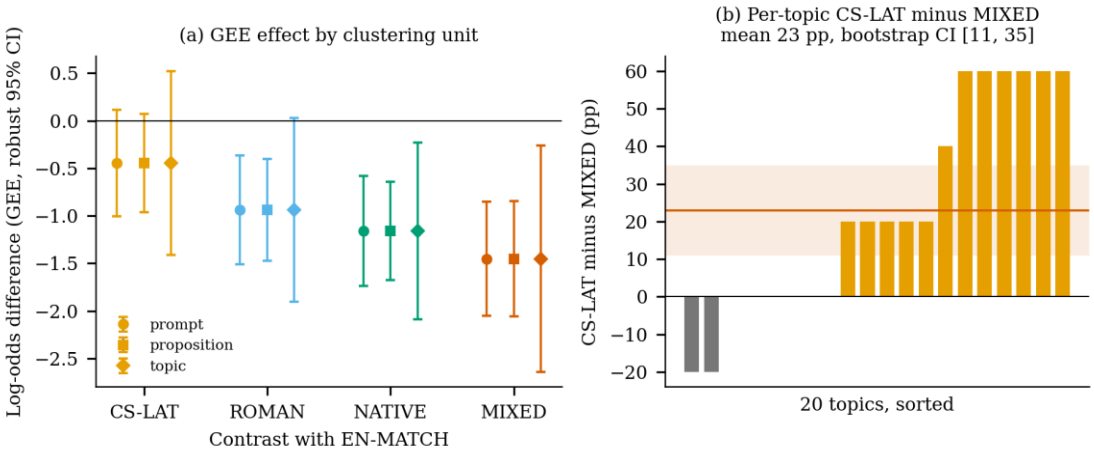


Fig. 4. (a) GEE log-odds differences from EN-MATCH (J-QWEN) with robust 95% CIs at three clustering levels (circle: prompt; square: proposition; diamond: topic). Intervals widen at topic level, where the CS-LAT and ROMAN intervals include zero. (b) Per-topic accuracy difference, CS-LAT minus MIXED (20 topics; band: bootstrap 95% CI of the mean; line: mean).

D. Languages and models

Accuracy by language is in Fig. 3 and Appendix Table X. MIXED is below CS-LAT in all five languages; EN-MATCH is above CS-LAT in four and tied with it in Bengali. A permutation test of whether the condition profile depends on language (statistic: interaction sum of squares of the language × condition accuracy table; null: language labels permuted among the five propositions of each topic; 10,000 permutations) gives $p = 0.17$. With 20 observations per cell this test has little power, so we report no detectable language dependence, not that languages behave alike; Bengali and Telugu differ visibly in the ROMAN column (20% vs. 50%), and nothing here can separate chance from effect.

ALLaM-2-7B answers 10% of matched-English questions correctly and 5%, 2%, 2% and 3% in CS-LAT, ROMAN, NATIVE and MIXED (stored labels), so the representation contrast cannot be measured on it. The five-condition rank correlation with Qwen3.8-27B under the stored labels ($\rho = 0.67$, $p = 0.22$) describes a floor, not a replication. We make no claim about other model families.

VI. TOKENIZATION AND THE OUTPUT BUDGET

A. Input tokenization (exploratory)

Subword fragmentation is a standard explanation for weaker performance on non-Latin scripts. Our earlier analysis measured characters per token with the Qwen2.5-7B tokenizer, which does not match the served model: it disagrees with the API's prompt-token counts and would put capped 128-token responses at 120–305 tokens. With the Qwen3.8-27B tokenizer, which does match (Section IV), Brahmic text is less fragmented than that analysis implied: prompts average 2.43 characters per token in NATIVE and 3.70 in MIXED, against 4.65 in EN-MATCH and 4.11 in CS-LAT (previously 1.24 and 2.49 for NATIVE and MIXED). Within the CS-LAT–MIXED contrast, MIXED lowers characters per token ($a = -0.41$, SE 0.10), but once condition is controlled the mediator's association with correctness is not significant and has the wrong sign for a fragmentation account ($b = -0.40$, SE 0.23, $p = 0.08$; 95% CI −0.85 to 0.05; Sobel $p = 0.11$). The implied indirect effect is positive while the total effect is negative (−1.01). Because characters per token is almost determined by condition and language in this design, the test has little leverage: it is exploratory and does not rule out a tokenization effect. Across the six conditions, characters per token and accuracy are rank-correlated ($\rho = 0.83$), but six points that differ in language frame, script and answer length identify nothing (Fig. 5c).

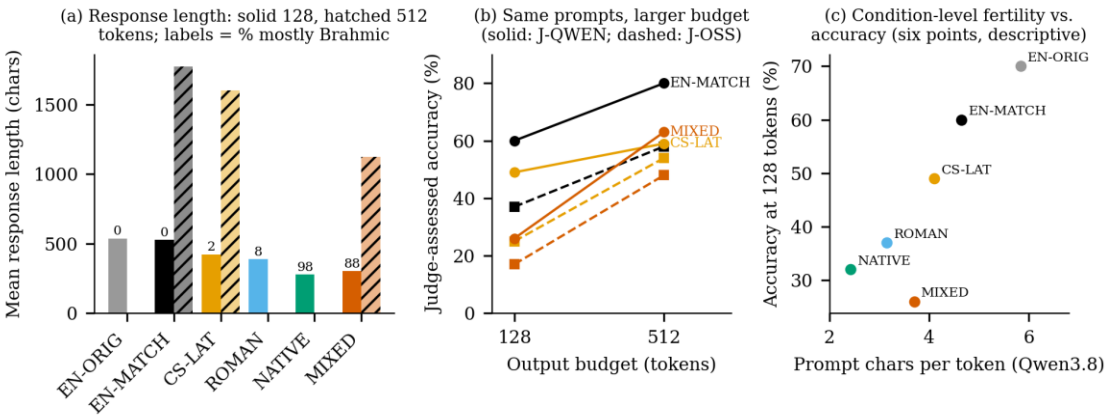


Fig. 5. (a) Mean response length at 128 tokens (solid; labels: % of responses written mostly in Brahmic script) and, for three conditions, at 512 tokens (hatched). (b) Accuracy of the same 300 prompts at 128 and 512 tokens, graded by J-QWEN (solid) and J-OSS (dashed). (c) Condition-level prompt characters per token (Qwen3.8 tokenizer) against accuracy at 128 tokens (six points; descriptive only).

B. Response script, length and the output budget

The prompt script largely determines the response script: 98% of NATIVE and 88% of MIXED responses are written mostly in Brahmic script (more than half of their letters), against 8% and 2% for ROMAN and CS-LAT and none for English. Brahmic answers cost more tokens per character (2.2–2.4 characters per token in NATIVE and MIXED responses against 4.1 in English ones), and nearly all responses used the full 128 tokens, so mean response length is 529 characters in EN-MATCH but 279 in NATIVE and 303 in MIXED (Fig. 5a). Among errors, the share whose judge reason mentions an incomplete or missing answer (a keyword rule, Appendix C) is 0 of 40 in EN-MATCH, 11 of 51 in CS-LAT, 36 of 68 in NATIVE and 38 of 74 in MIXED. The rule depends on the judge's wording and does not measure truncation, but it points the same way.

To separate script from budget we re-ran EN-MATCH, CS-LAT and MIXED (300 prompts) with a 512-token limit and nothing else changed: same model identifier, system prompt, decoding settings and prompts. Both the 128-token and the 512-token responses were graded by J-QWEN and by J-OSS, so every comparison across budgets uses one judge. The limit still binds for most code-switched answers (58 of 100 EN-MATCH, 15 CS-LAT and 5 MIXED responses stopped before 512 tokens), but MIXED responses now average 1,122 characters, twice what English responses had at 128 tokens. Table IV gives the result. Under J-QWEN, MIXED rises from 26% to 63% and CS-LAT from 49% to 59%, so the MIXED–CS-LAT gap moves from −23 to +4 points ($p = 0.57$ at 512 tokens; change in the gap +27 points, topic-bootstrap CI [7, 45], permutation $p = 0.018$). Under J-OSS the gap is −8 points at 128 tokens and −6 at 512, neither significant. On the 60 word-identical propositions the picture is the same (J-QWEN, 512 tokens: 62% vs. 63%). MIXED with 512 tokens matches EN-MATCH with 128 (63% vs. 60%, $p = 0.77$). The English–CS-LAT gap behaves differently: under J-QWEN it widens to 21 points at 512 tokens ($p < 0.001$; topic-level $p = 0.082$), while under J-OSS it shrinks to 4 points ($p = 0.57$).

TABLE IV
Output-budget follow-up. The same 300 prompts answered with 128 and 512 tokens; both budgets graded by the same judge. Accuracy (%), and gaps in points with exact McNemar p / topic-level permutation p.

	J-QWEN 128	**J-QWEN 512**	**J-OSS 128**	J-OSS 512
EN-MATCH	60	80	37	58
CS-LAT	49	59	25	54
MIXED	26	63	17	48
CS-LAT − EN-MATCH	−11 (.13/.42)	−21 (<.001/.08)	−12 (.05/.22)	−4 (.57/.76)
MIXED − CS-LAT	−23 (<.001/.004)	+4 (.57/.65)	−8 (.18/.24)	−6 (.34/.44)
Stopped before cap	0, 1, 0	58, 15, 5	0, 1, 0	58, 15, 5

Three conclusions follow. First, the 128-token MIXED penalty is not evidence that script alternation harms recall: given room, the model states the gold fact about equally often in the mixed-script and Latin-script code-switched conditions. Its most likely source is that Brahmic-script answers reach the cap before reaching the fact, and answers a human would call incomplete are then graded as wrong; the EN-MATCH answer on the MTP Act, for instance, correctly states the one-practitioner rule for the first 20 weeks, is cut off before the 20–24-week rule, and is graded HALLUCINATED. Second, NATIVE and MIXED behave alike at 128 tokens (32% and 26%, $p = 0.29$) although MIXED has five script boundaries per prompt and NATIVE one, which also argues against a boundary-density account. Third, the English advantage is unresolved: it persists under one judge and nearly vanishes under the other, which is a question about the judge (Section VII) as much as about the model.

VII. JUDGE DEPENDENCE

The judge matters in three ways. *Configuration.* Our first run graded the Indic conditions at temperature 0.3 with the answering system prompt and EN-MATCH at temperature 0 with an evaluator prompt. Harmonising the configuration changed 6–10% of the labels in each re-graded condition and moved CS-LAT accuracy from 43% to 49%, which is why the stored English–CS-LAT gap (17 points) exceeded the harmonised one (11). *Family.* J-OSS, which did not produce the answers, is far stricter than the self-judge: on the 128-token outputs it accepts 37% of EN-MATCH answers against 60% for J-QWEN, agrees with J-QWEN on 76% of items (κ = 0.33–0.62 by condition; Table V), and 128 of the 146 disagreements are answers J-QWEN accepts and J-OSS rejects. Inspection shows why: J-QWEN often accepts an answer that is on topic and accurate but does not state the gold fact (for example, one that only names the governing Act), whereas J-OSS requires the gold fact. The rubric ("judge only factual correctness and whether it actually addresses the question") does not settle which is right, and truncated answers make the case common. *Differential error.* Neither judge has been compared with human labels, so we cannot say which is closer to a careful reader or whether either is harsher on one script. The two agree more often on NATIVE, ROMAN and MIXED (79–85%) than on English and CS-LAT (64–73%), so their disagreement is not concentrated on Brahmic text.

TABLE V

Judge agreement on the 600 outputs at 128 tokens. Stored vs. J-QWEN: share of identical labels (EN-MATCH was stored under the J-QWEN configuration). J-QWEN vs. J-OSS: share identical and Cohen's κ.

Condition	Stored vs. J-QWEN	J-QWEN vs. J-OSS	κ
EN-ORIG	0.94	0.64	0.33
EN-MATCH	1.00	0.73	0.49
CS-LAT	0.90	0.72	0.43
ROMAN	0.94	0.81	0.56
NATIVE	0.92	0.85	0.62
MIXED	0.92	0.79	0.39

Table VI turns the remaining uncertainty into tipping points. Assuming English answers are judged without error, it gives the sensitivity the judge would need on the comparison condition for that condition's true accuracy to equal the reference's. A correction applied equally to both conditions (the Rogan–Gladen estimator [22]) divides a gap by TPR − FPR and can only enlarge it, so a common-error grid cannot show robustness; the relevant threat is differential error. At a false-accept rate of 0.08, erasing the CS-LAT gap would need a judge that rejects 24% of truly correct CS-LAT answers (sensitivity 0.76), and erasing the MIXED gap one that rejects 62% (0.38). These are hypothetical error rates, not estimates of the judge's actual errors, which are unknown; the cross-family disagreement shows they are not small. A crude check agrees: on the 330 outputs whose gold answer contains numerals, the J-QWEN label matches whether all gold numerals appear in the response 67% of the time, with disagreements in both directions (73 accepted answers lack the figure; 35 rejected ones contain it).

TABLE VI

Sensitivity the judge would need on the comparison condition for its true accuracy to equal the reference condition's, at the given false-accept rate (J-QWEN, 128 tokens). Reference answers assumed judged without error. Lower entries mean a larger error would be needed.

0.00	**0.82**	**0.43**	**0.53**
0.04	0.79	0.41	0.49
0.08	0.76	0.38	0.45
0.12	0.74	0.35	0.41
0.16	0.71	0.33	0.36

VIII. DISCUSSION

What the data support is narrower than the design promised. For this model, these 20 topics and a 128-token budget, restating a question in Romanized or Brahmic script, or as code-switched text, lowers the share of answers a judge accepts, and mixed script is lowest. Quadrupling the budget removes the mixed-script penalty relative to Latin-script code-switching under both judges, and MIXED then does as well as English did at 128 tokens. The parsimonious reading is that much of the representation effect at 128 tokens reflects how much answer fits into the budget in the response script, not comprehension of the prompt. Whether an English advantage remains once answers can be completed depends on the judge, and topic-level evidence for it is weak in every analysis.

The practical lesson is methodological. Output caps are often set low to save cost, and tokenizers make some scripts more expensive per character; a shared cap then gives answers in those scripts less room. Studies that compare languages or scripts through generated answers should report how often the cap binds in each condition and check that conclusions survive a larger cap; judges should be configured identically across conditions, and at least one should come from a family other than the evaluated model's. Table VII lists claims we examined and could not support.

TABLE VII

Claims examined and not supported.

Examined claim	Outcome
Script alternation (CS-LAT → MIXED) lowers recall	Not supported: gap +4 at 512 tokens ($p = 0.57$); n.s. under J-OSS
English advantage over CS-LAT at topic level	Not supported (topic $p \geq 0.08$ in all analyses)
Prompt token fertility mediates CS-LAT → MIXED	Not detected ($p = 0.08$, wrong sign; exploratory)
Language dependence of the profile	Not detected ($p = 0.17$), low power
Rank agreement with ALLaM-2-7B	Uninformative (floor; ρ = 0.67, $p = 0.22$)
Errors rise with script transitions	Not supported (NATIVE ≈ MIXED)
Original 20-fold truncation ratio	Not reproducible from files

IX. LIMITATIONS

Topics. Twenty topics with heavy within-topic concentration bound what any statistic can show (effective n ≈ 67 of 500). A 25-proposition expansion has been written but not evaluated; its design power (Appendix D) is a projection.

Output budget. Even 512 tokens bound most code-switched answers; we have not run an uncapped condition or one that fixes the answer language, and we re-ran three of the six conditions. **Judges.** Two LLM judges that disagree on a quarter of items, and no human labels on any output; a stratified, blinded sample of 210 outputs and an annotation protocol are prepared but not yet annotated. **Questions and gold answers.** One generic template that under-specifies the target for several topics; gold answers checked by the author and volunteers without an item-level record; one superseded scheme. **Prompts.** No fluent-speaker validation of naturalness or equivalence; CS-LAT and MIXED differ by one word in Telugu and Kannada, and Hindi ROMAN is not a transliteration of Hindi NATIVE. **Models and languages.** One informative model, whose served weights we cannot verify, and five languages. **Single run.** No repeated sampling; hosted greedy decoding was not fully deterministic, and API behaviour can change over time.

X. ETHICS AND BROADER IMPACT

Evaluations like this one guide where civic and legal information systems are considered safe to deploy for speakers who write the way our conditions do. Overstated gaps could discourage building for these users; understated ones could hide failures. We therefore report effect sizes with their dependence on clustering, output budget and judge. The prompts contain no personal data. The benchmark contains legal and policy facts that become outdated (one already has) and should not be used as legal advice. We do not estimate how many people use each representation, and make no claim about it.

REPRODUCIBILITY

All numbers, tables and figures are produced from the raw files by `paper_v2/scripts/make_all.py` (128-token analyses; `--labels jqwen` for the harmonised labels), `analyze_exp004.py` (re-grading and budget follow-up) and `make_figures.py` (Appendix E). The follow-up cost $1.10 at the provider's list prices in October 2026. The project's budget ledger records $0.21 for all earlier runs, but it priced every call at a flat $0.08 per million tokens; at current list prices the Qwen3.8-27B generation and grading in EXP-002 and EXP-003 alone would cost about $2.1. The provider's billing record is the authoritative figure.

APPENDIX A: BENCHMARK DETAILS

A.1. Topics

Table VIII lists the 20 topics, the provision against which each gold answer was reviewed, the outcome of that review, and the number of languages (of five) in which each condition was answered correctly under J-QWEN. Topics are Indian public-policy and regulatory subjects, several comparing two schemes or institutions (e.g. NEFT vs. RTGS, Covaxin vs. Covishield, PSLV vs. GSLV Mk III); not all are statutes, so we call them public-policy topics. Gold column: *ok*, consistent and adequate; *simpl.*, a correct simplification; *narrow*, other correct answers to the generic question exist; *dated*, the scheme has been superseded.

TABLE VIII
Topics, provision used to review the gold answer, review outcome, and languages (of five) answered correctly per condition (Qwen3.8-27B, J-QWEN, 128 tokens).

Topic	Provision reviewed	Gold	EN-ORIG	EN-MATCH	CS-LAT	ROMAN	NATIVE	MIXED
Citizenship Amendment Act Cut-off	Citizenship (Amendment) Act 2019, s. 2	ok	5	5	1	3	4	2
Companies Act CSR Mandate	Companies Act 2013, s. 135(1)	ok	5	5	3	2	0	0
Covaxin vs Covishield	CDSCO approvals, Jan. 2021	ok	5	5	5	4	2	4
GST Registration Exemption	CGST Act s. 22; Notif. 10/2019-CT	narrow	5	5	4	2	4	1
ISRO PSLV vs GSLV Mk III	ISRO launcher specifications	ok	5	5	2	0	1	0
Kharif vs Rabi Rice	ICAR crop-season references	simpl.	5	5	5	4	4	4
Lok Sabha vs Rajya Sabha Money Bill	Constitution, Arts. 109–110	ok	5	5	2	3	3	3
National Park vs Wildlife Sanctuary	Wild Life (Protection) Act 1972, ss. 33, 35	ok	5	5	3	2	2	0
Patents Act Compulsory License	Patents Act 1970, s. 84(1)	ok	5	5	3	3	3	0
SEBI Insider Trading Pre-Clearance	SEBI PIT Regulations 2015, Sch. B	ok	5	5	4	2	1	1
Arbitration Act Time Limit	Arbitration Act 1996, s. 29A	ok	0	5	1	0	0	0
PMFBY vs WBCIS	PMFBY/RWBCIS operational guidelines	ok	0	5	0	0	0	0
Classical Tamil vs Sanskrit Grammar	Philological references	ok	5	0	3	1	0	3
Medical Termination of Pregnancy Act	MTP Act 1971 (am. 2021), s. 3(2)	narrow	5	0	0	1	0	0
NEFT vs RTGS	RBI NEFT/RTGS directions	ok	5	0	4	5	4	4
RTI Third Party Information	RTI Act 2005, s. 11(1)	ok	5	0	1	0	0	0
Environment Clearance Public Hearing	EIA Notification 2006, para. 7	narrow	0	0	0	0	0	0
IBC Section 7 Default Threshold	IBC 2016, s. 4 (₹1 crore from 2020)	ok	0	0	0	0	0	0
PM-KISAN vs Rythu Bandhu	PM-KISAN guidelines; Telangana G.O.s	dated	0	0	3	2	1	0
Supreme Court vs High Court Writ Jurisdiction	Constitution, Arts. 32, 226	ok	0	0	5	3	3	4

A.2. Partitions and the unevaluated expansion

Version 1.1-CANDIDATE: development 1,000 groups, validation 200, in-distribution test 200, out-of-distribution test 100 (the evaluated core). The 200 in-distribution test groups were run in EXP-002 for both models; they use fictitious entities, and 2 of 2000 responses were judged correct, so they are not informative about recall. The prospective 1.2-PILOT adds 25 propositions (125 prompts) on topics with zero entity overlap with the evaluated 20; no model has been run on them.

APPENDIX B: ADDITIONAL RESULTS

Excluding the three topics with narrow gold answers (GST, MTP Act, Environment Clearance; 85 propositions), the J-QWEN differences from EN-MATCH are −12 (CS-LAT, p = 0.16), −25 (ROMAN, p = 0.002), −32 (NATIVE, p < 0.001) and −35 points (MIXED, p < 0.001), and MIXED vs. CS-LAT is −23.5 points (p < 0.001; topic-level p = 0.007).

TABLE IX

All pairwise contrasts, J-QWEN labels, 128 tokens (proposition-level exact McNemar; b: the second-named (reference) condition correct and the first wrong; c: the reverse; topic-level permutation p and bootstrap CI for the change, uncorrected). The first five rows are the primary family.

CS-LAT vs. EN-MATCH	**−11**	**27**	**16**	**0.13**	**0.42**	**[−34, 13]**
ROMAN vs. EN-MATCH	−23	35	12	0.001	0.09	[−46, 0]
NATIVE vs. EN-MATCH	−28	36	8	2.54×10^{-5}	0.03	[−49, −6]
MIXED vs. EN-MATCH	−34	45	11	5.38×10^{-6}	0.03	[−59, −7]
MIXED vs. CS-LAT	−23	30	7	1.91×10^{-4}	0.004	[−35, −11]
EN-MATCH vs. EN-ORIG	−10	20	10	0.10	0.69	[−35, 15]
MIXED vs. ROMAN	−11	19	8	0.05	0.07	[−21, −1]
MIXED vs. NATIVE	−6	14	8	0.29	0.46	[−18, 6]
ROMAN vs. CS-LAT	−12	25	13	0.07	0.06	[−22, −2]
NATIVE vs. CS-LAT	−17	27	10	0.008	0.03	[−30, −4]
NATIVE vs. ROMAN	−5	16	11	0.44	0.37	[−13, 3]

TABLE X

Correct answers by language and condition, as percentages (J-QWEN; n = 20 per cell).

Hindi	**70**	**60**	**45**	**40**	**40**	**30**
Bengali	70	60	60	20	20	35
Tamil	70	60	50	35	35	15
Telugu	70	60	40	50	30	35
Kannada	70	60	50	40	35	15
All	70	60	49	37	32	26

TABLE XI

Correct answers out of 100 per model and condition, 128 tokens. ALLaM-2-7B has stored labels only; its Indic conditions were graded under the EXP-002 judge setting and EN-MATCH under the EXP-003 setting.

Condition	**Qwen stored**	**Qwen J-QWEN**	**ALLaM-2-7B**	ALLaM 95% CI
EN-ORIG	64	70	8	[4.1, 15.0]
EN-MATCH	60	60	10	[5.5, 17.4]
CS-LAT	43	49	5	[2.2, 11.2]
ROMAN	33	37	2	[0.6, 7.0]
NATIVE	28	32	2	[0.6, 7.0]
MIXED	24	26	3	[1.0, 8.5]

TABLE XII

Response- and prompt-side descriptors by condition (Qwen3.8-27B, 128 tokens). Script transitions: Latin ↔ Brahmic changes between adjacent words in the prompt. Chars/tok: prompt characters per token, Qwen3.8-27B tokenizer. Brahmic: % of responses mostly in Brahmic script. Errors and Incompl.: J-QWEN labels and reasons; Incompl. is the keyword rule of Appendix C.

Condition	**Script trans.**	**Chars/tok**	**Resp. chars**	**Brahmic (%)**	**Errors**	**Incompl.**
EN-ORIG	0.0	5.83	538	0	30	0
EN-MATCH	0.0	4.65	529	0	40	0
CS-LAT	0.0	4.11	423	2	51	11
ROMAN	0.0	3.15	390	8	63	18
NATIVE	1.0	2.43	279	98	68	36
MIXED	5.0	3.70	303	88	74	38

APPENDIX C: RULES AND RUBRICS

Incompleteness rule. A response is flagged when (i) it used the full 128 completion tokens, (ii) its last non-space character is not a sentence-final mark, (iii) the judge labelled it incorrect, and (iv) the judge's reason matches one of *incomplete*, *truncated*, *cut off*, *abrupt*, *unfinished*, *missing specific*, *missing parameter*, *fails to finish*. Condition (iv) depends on the judge's wording; the rule is a descriptive device, not a measure of truncation. **Judge prompt.** "You are grading whether a model's ANSWER to a user QUESTION is factually correct. You are given a trusted GOLD answer. The answer may mix an Indian language with English (code-switching) and may be phrased very differently from the gold—that is fine. Judge ONLY factual correctness and whether it actually addresses the question." Output: `VERDICT: <CORRECT or HALLUCINATED> | REASON: <max 12 words>`. **Judge settings.** Stored, EXP-002: Qwen3.8-27B, temperature 0.3, at most 256 tokens, system prompt "You are a helpful assistant answering questions from Indian users. Questions may mix an Indian language with English (code-switching).

Answer accurately and concisely in the same mixed style the user used." Stored, EXP-003, and J-QWEN: temperature 0, top-p 1, at most 256 tokens, system prompt "You are an expert factuality evaluator." J-OSS: gpt-oss-120b, temperature 0, top-p 1, low reasoning effort, at most 1,024 tokens, same system prompt. All 1,800 follow-up judgements were parsed. **Script transitions.** Words are maximal letter runs, each labelled Latin or Brahmic (U+0900–U+0DFF); transitions are label changes between consecutive words, giving 5.0 per MIXED prompt, 1.0 per NATIVE prompt (the Latin topic string) and 0 elsewhere. **Mostly Brahmic.** More than half of a response's letters and combining marks lie in U+0900–U+0DFF. **Templates.** All 20 Indic templates are listed in the released prompt-validation protocol.

APPENDIX D: PROSPECTIVE DESIGN CALCULATION

A normal-approximation calculation for a 45-topic design (225 paired propositions, design effect $1 + (5 - 1)\rho$, binary variance at $\bar{p} = 0.4$, two-sided $\alpha = 0.05$) with $\rho = 0.2663$ and a 21-point effect gives power 88.6% (minimum detectable effect 18.6 points); the same formula gives 55.9% for the 20 evaluated topics. Those parameters come from the first English template and the stored labels. With the harmonised labels ($\rho = 0.27$) the 11-point English–CS-LAT gap would have power 38% with 45 topics and 19% with 20, and a 23-point gap 93% and 63%. The formula ignores pairing and is a rough projection; the 25 new propositions have not been evaluated.

APPENDIX E: REPRODUCIBILITY

Python 3.11 with numpy, pandas, scipy, statsmodels, tokenizers and matplotlib. From the repository root, `python paper_v2/scripts/make_all.py` (stored labels) and `make_all.py --labels jqwen` read the two prediction files and write `paper_v2/output/numbers_*.json`; `python scripts/run_exp004_budget_rejudge.py --stage gen|judge` produced the follow-up (Groq key required; outputs, finish reasons and a spend log in `results/EXP-004-budget-rejudge/`); `analyze_exp004.py` and `make_figures.py` write the remaining numbers and Figs. 2–5. The tokenizer check downloads the Qwen/Qwen2.5-7B and Qwen/Qwen3.8-27B tokenizers. Sign-flip tests enumerate all 2^{20} patterns; bootstraps and the language permutation use fixed seeds. Topic-level p values in an earlier draft were Monte Carlo approximations and differ from the exact values here in the third decimal. That draft's language-dependence p value (0.15) and numeral check (60% agreement) could not be reproduced because the script that produced them was not in the repository; they are replaced by the procedures described above.

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